

```
%root_path = fileparts(mfilename("fullpath"));
%root_path = pwd;
scuts_path = 'e:\SHARE\Fe\Data\analysis\07_fit_with_multicut_allJ\sel_cuts';
Cutfitfile = fullfile(scuts_path, 'NG_FFCutsFits.mat');
```

```
if ~exist('NG_FFCutsFits', 'var')
    ld = load(Cutfitfile);
    NG_FFCutsFits = ld.fit_src_struct;
end
```

```
fn = fieldnames(NG_FFCutsFits);
nf = 11
```

```
nf =
11
```

```
%for i = 11:numel(fn)
%selectCuts = fn{nf};
selectCuts = 'Ei200_off200NGmaxFF'
```

```
selectCuts =
'Ei200_off200NGmaxFF'
```

```
if contains(selectCuts, 'msk')
    error('Can not fit field: %s', selectCuts);
end
if contains(selectCuts, 'all_fit_par')
    error('Can not fit field: %s', selectCuts);
end

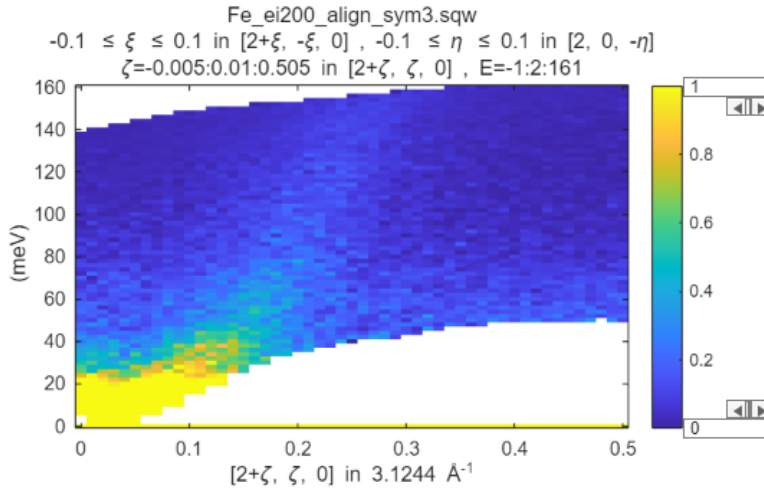
fprintf('Processing cuts: %s \n', selectCuts);
```

```
Processing cuts: Ei200_off200NGmaxFF
```

```
%selectCuts= 'Ei400_off200GPmaxFF';
cuts_list = NG_FFCutsFits.(selectCuts);
if ~iscell(cuts_list)
    cuts_list = {cuts_list};
end
if contains(selectCuts, 'Ei200')
    mi = maps_instrument(200, 600, 'S');
end
if contains(selectCuts, 'Ei400')
    mi = maps_instrument(401, 600, 'S');
end
if contains(selectCuts, 'Ei800')
    mi = maps_instrument(786, 600, 'S');
end
sample=IX_sample(true, [1,0,0], [0,1,0], 'cuboid', [0.04,0.03,0.02]);
for j=1:numel(cuts_list)
```

```
cuts_list{j} = cuts_list{j}.set_instrument(mi);
cuts_list{j} = cuts_list{j}.set_sample(sample);
plot(cuts_list{j}); lz 0 1; keep_figure;
```

end



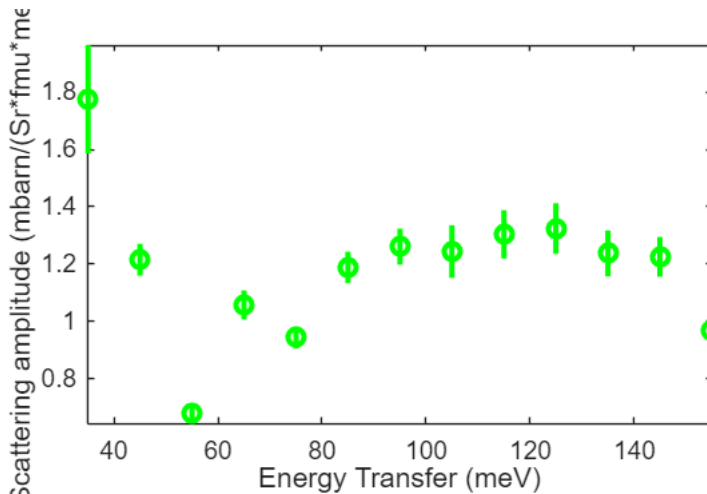
```
dir_name = "NG";

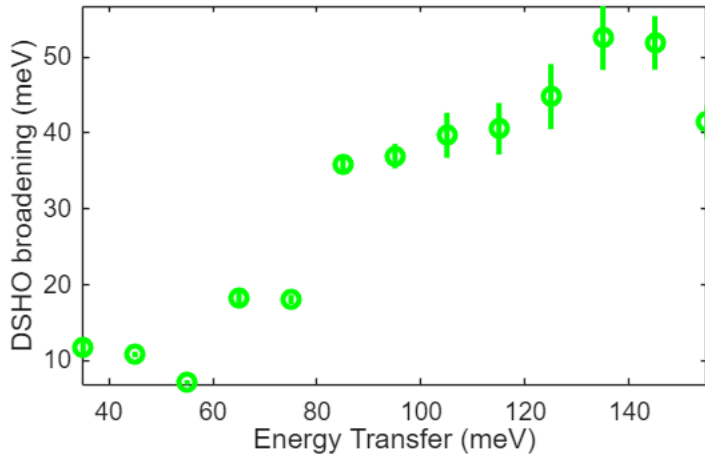
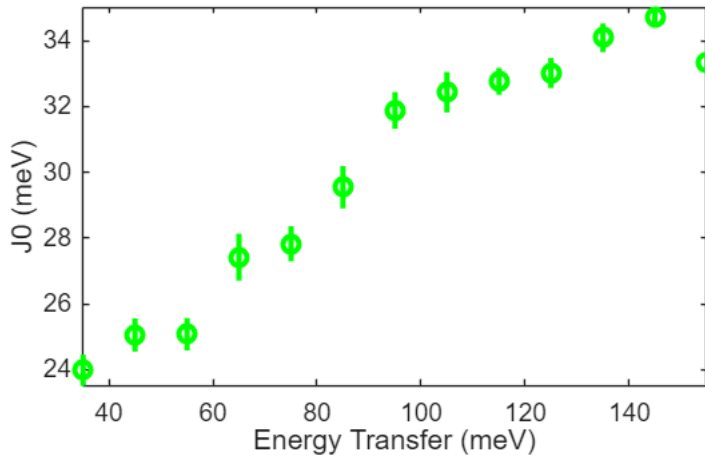
dE_step = 2; %original energy transfer step data were binned to. No point in going
finer
half_dE = 5; % half width of data binning

en_mm = cuts_list{1}.data.img_range(:,4);
en_mm(1) = max(30,en_mm(1));
%en_mm(1) = 100;
%en_mm(2) = 170;
cut_en = en_mm(1)+half_dE:2*half_dE:(en_mm(2)-half_dE);

cut_dimensions = 1;
NG_FFCutsFits.(selectCuts) = cuts_list;

NG_FFCutsFits = fit_multicutsFF_along_direction(...
    NG_FFCutsFits,selectCuts,cut_dimensions,cut_en,dE_step,half_dE,true);
```

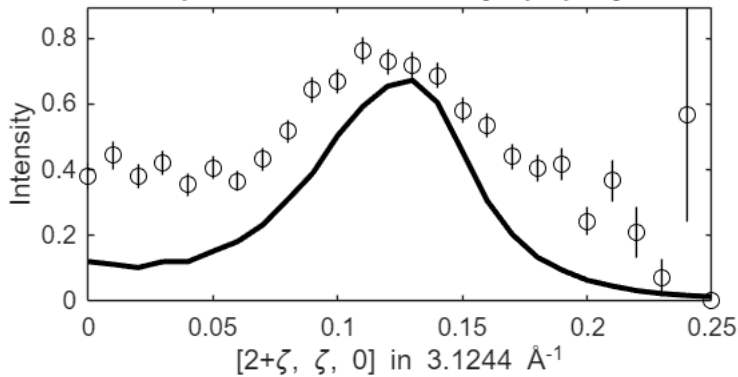




 **** En = 35±5 going to 155

Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 1.9608

Fe_ei200_align_sym3.sqw
 $\xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 30 :
 $\zeta = -0.005:0.01:0.505$ in $[2+\zeta, \zeta, 0]$

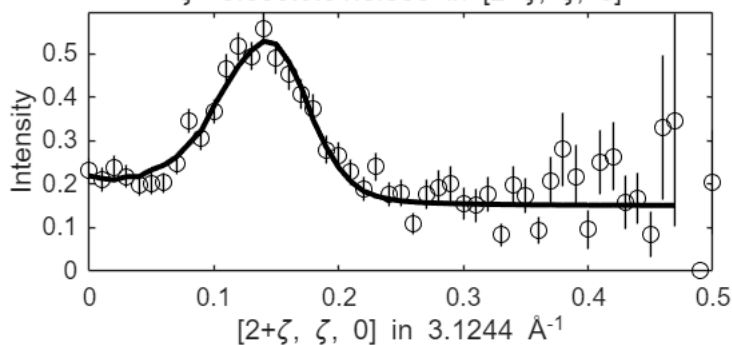


 **** En = 45±5 going to 155

Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 1.9608

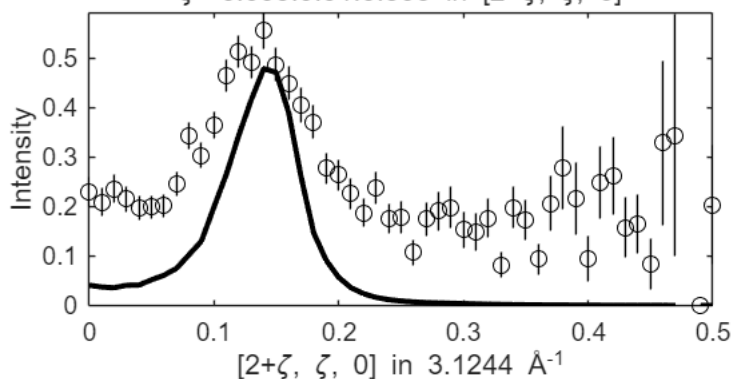
Fe_ei200_align_sym3.sqw

1.2 ± 0.054 ; $J_0 = 25 \pm 0.5$; $J_1 = 0 \pm 0$; $\gamma = 10.4583 \pm 0.3$
 $\xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 40
 $\zeta = -0.005:0.01:0.505$ in $[2+\zeta, \zeta, 0]$



Fe_ei200_align_sym3.sqw

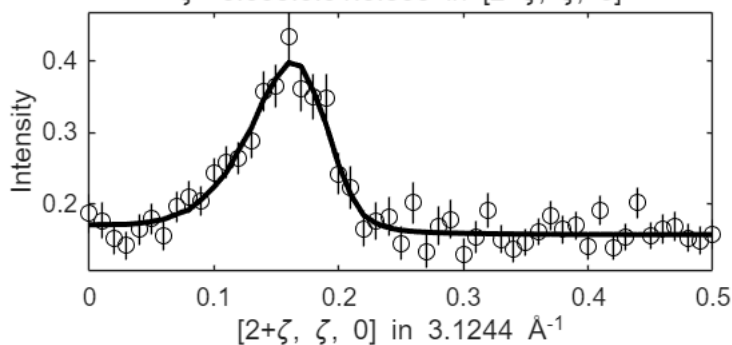
$\xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 40
 $\zeta = -0.005:0.01:0.505$ in $[2+\zeta, \zeta, 0]$



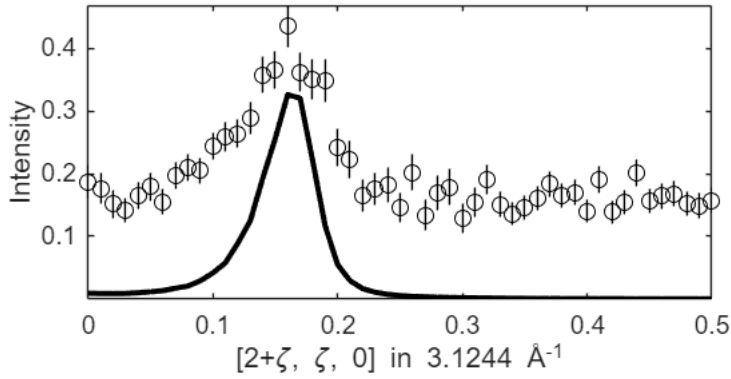
**** En = 55±5 going to 155

Fe_ei200_align_sym3.sqw

$.68 \pm 0.029$; $J_0 = 25 \pm 0.5$; $J_1 = 0 \pm 0$; $\gamma = 7.16052 \pm 0.2$
 $\xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 50
 $\zeta = -0.005:0.01:0.505$ in $[2+\zeta, \zeta, 0]$

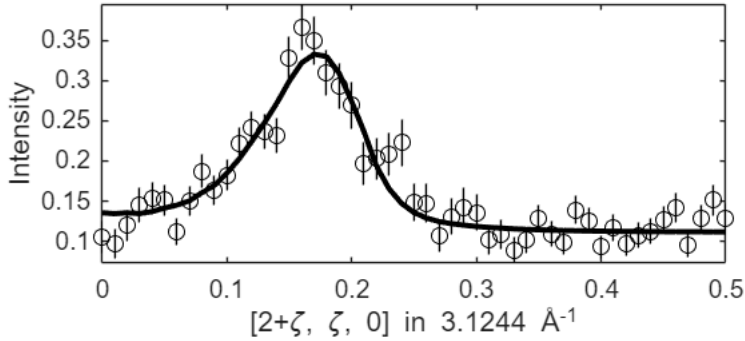


Fe_ei200_align_sym3.sqw
 $\xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 50
 $\zeta = -0.005:0.01:0.505$ in $[2+\zeta, \zeta, 0]$

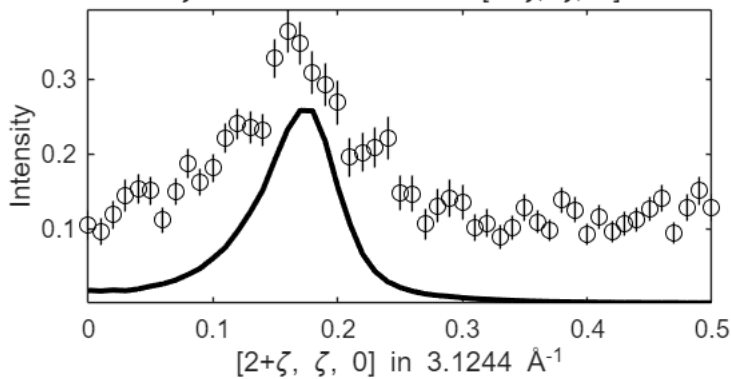


 **** En = 65±5 going to 155

Fe_ei200_align_sym3.sqw
 1.1 ± 0.052 ; $J_0 = 27 \pm 0.73$; $J_1 = 0 \pm 0$; $\gamma = 18.2612 \pm 0.6$
 $\xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 60
 $\zeta = -0.005:0.01:0.505$ in $[2+\zeta, \zeta, 0]$



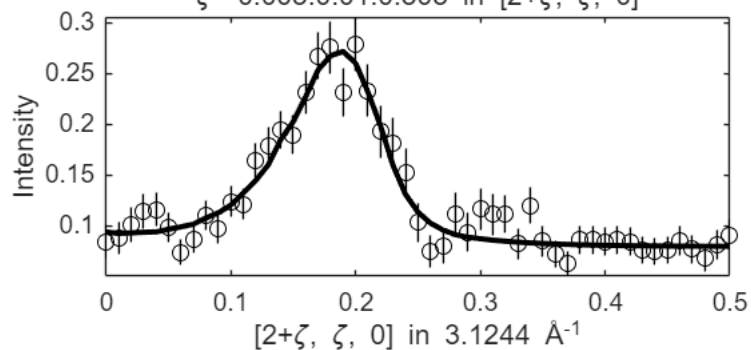
Fe_ei200_align_sym3.sqw
 $\xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 60
 $\zeta = -0.005:0.01:0.505$ in $[2+\zeta, \zeta, 0]$



 **** En = 75±5 going to 155

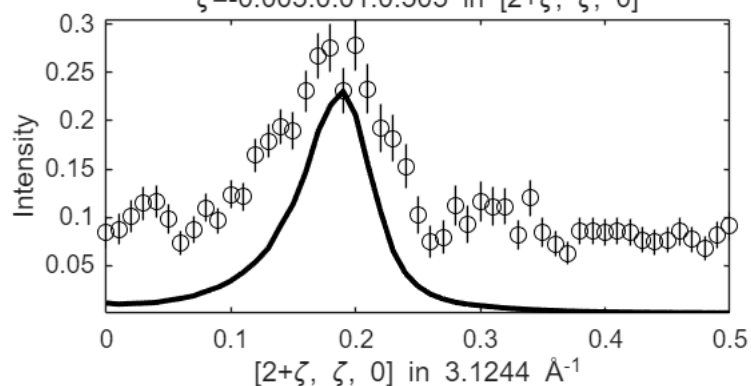
Fe_ei200_align_sym3.sqw

1.94 ± 0.082 ; $J_0 = 28 \pm 0.54$; $J_1 = 0 \pm 0$; $\gamma = 18.0204 \pm 2.0$
 $\xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 70
 $\zeta = -0.005:0.01:0.505$ in $[2+\zeta, \zeta, 0]$



Fe_ei200_align_sym3.sqw

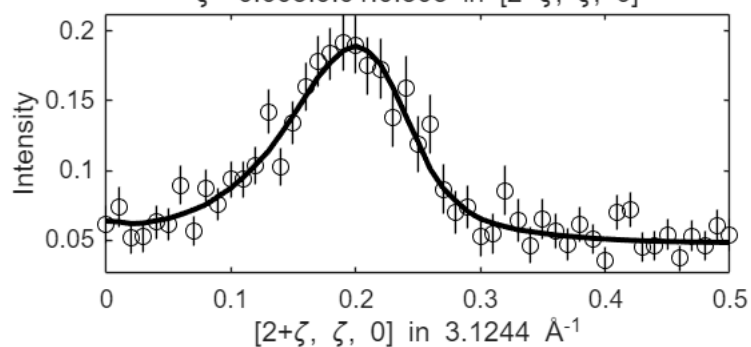
$\xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 70
 $\zeta = -0.005:0.01:0.505$ in $[2+\zeta, \zeta, 0]$



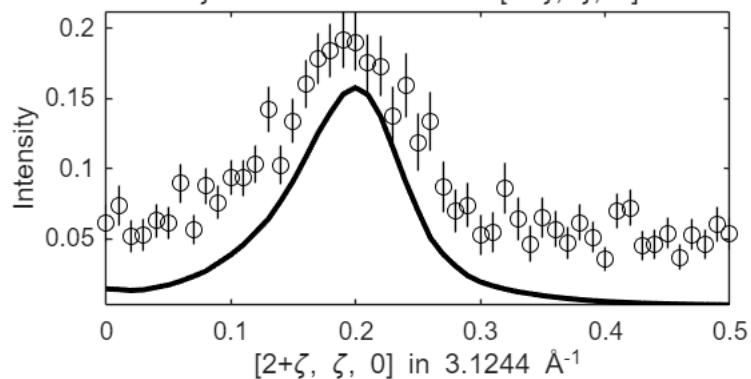
**** En = 85 ± 5 going to 155

Fe_ei200_align_sym3.sqw

1.2 ± 0.13 ; $J_0 = 30 \pm 0.79$; $J_1 = 0 \pm 0$; $\gamma = 35.9498 \pm 5.7$
 $\xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 80
 $\zeta = -0.005:0.01:0.505$ in $[2+\zeta, \zeta, 0]$

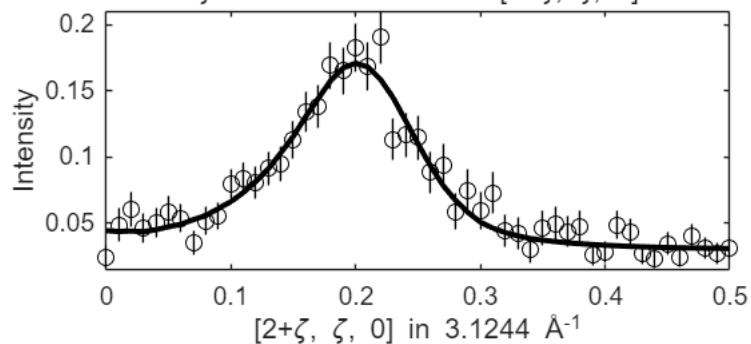


Fe_ei200_align_sym3.sqw
 $\xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, $80 \leq \zeta \leq 90$ in $[2+\zeta, \zeta, 0]$

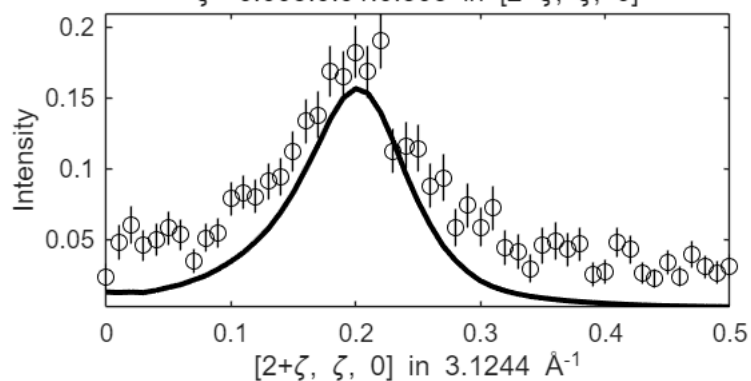


**** En = 95±5 going to 155

Fe_ei200_align_sym3.sqw
 1.3 ± 0.063 ; $J_0 = 32 \pm 0.55$; $J_1 = 0 \pm 0$; $\gamma = 37.0562 \pm 1.1$
 $\xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, $90 \leq \zeta \leq 95$ in $[2+\zeta, \zeta, 0]$

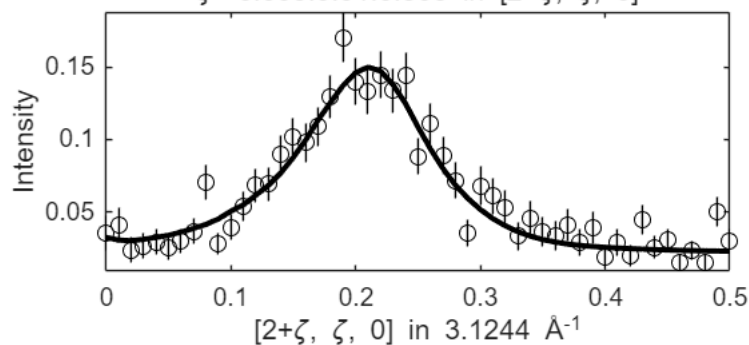


Fe_ei200_align_sym3.sqw
 $\xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, $90 \leq \zeta \leq 105$ in $[2+\zeta, \zeta, 0]$

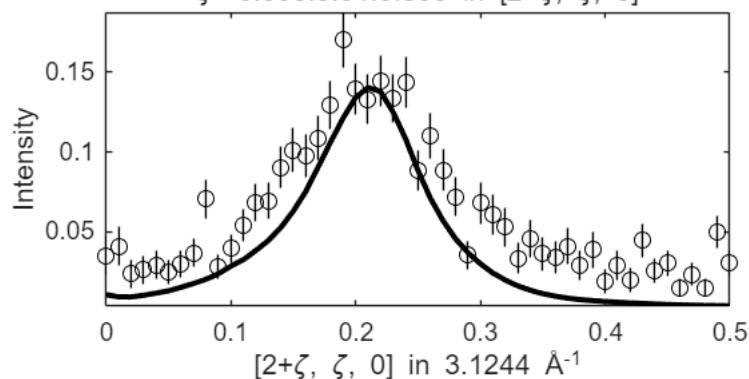


**** En = 105±5 going to 155

Fe_ei200_align_sym3.sqw
 1.2 ± 0.091 ; $J_0 = 32 \pm 0.61$; $J_1 = 0 \pm 0$; $\gamma = 39.6492 \pm 2.1$
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 100
 $\zeta = -0.005:0.01:0.505$ in $[2+\zeta, \zeta, 0]$

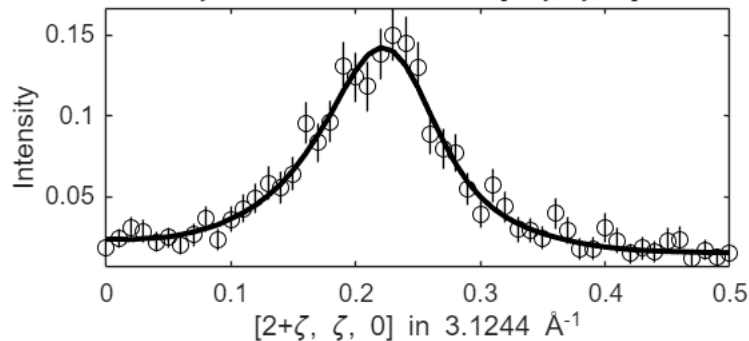


Fe_ei200_align_sym3.sqw
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 100
 $\zeta = -0.005:0.01:0.505$ in $[2+\zeta, \zeta, 0]$

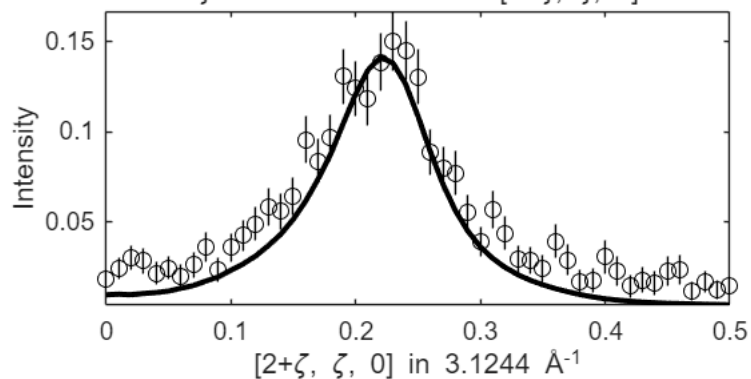


 **** En = 115 ± 5 going to 155

Fe_ei200_align_sym3.sqw
 1.3 ± 0.083 ; $J_0 = 33 \pm 0.41$; $J_1 = 0 \pm 0$; $\gamma = 40.5888 \pm 3.1$
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 110 :
 $\zeta = -0.005:0.01:0.505$ in $[2+\zeta, \zeta, 0]$

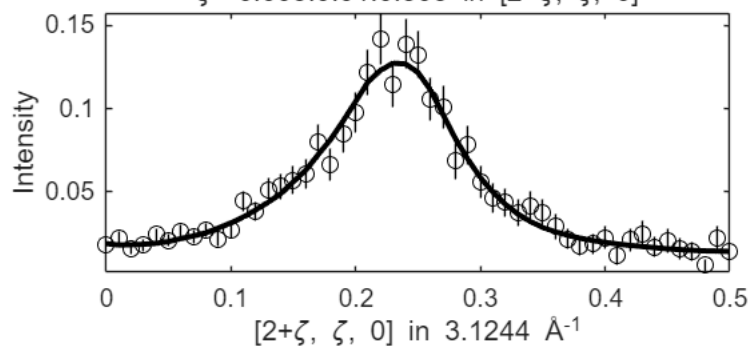


Fe_ei200_align_sym3.sqw
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 110 :
 $\zeta=-0.005:0.01:0.505$ in $[2+\zeta, \zeta, 0]$

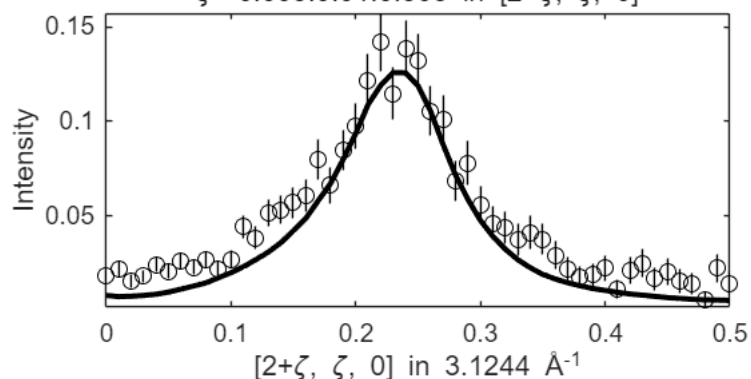


**** En = 125±5 going to 155

Fe_ei200_align_sym3.sqw
 1.3 ± 0.089 ; $J_0 = 33 \pm 0.44$; $J_1 = 0 \pm 0$; $\gamma = 44.7532 \pm 4.1$
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 120 :
 $\zeta=-0.005:0.01:0.505$ in $[2+\zeta, \zeta, 0]$



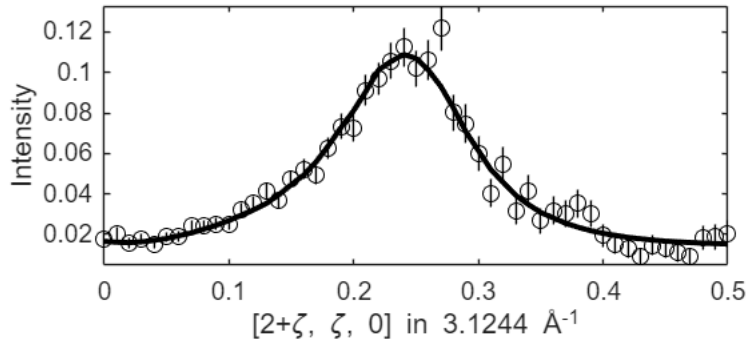
Fe_ei200_align_sym3.sqw
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 120 :
 $\zeta=-0.005:0.01:0.505$ in $[2+\zeta, \zeta, 0]$



**** En = 135±5 going to 155

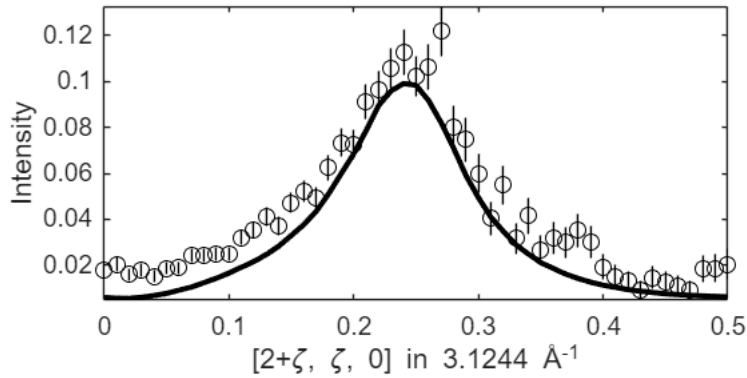
Fe_ei200_align_sym3.sqw

1.2 ± 0.082 ; $J_0 = 34 \pm 0.46$; $J_1 = 0 \pm 0$; $\gamma = 52.7299 \pm 4.7$
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 130
 $\zeta = -0.005:0.01:0.505$ in $[2+\zeta, \zeta, 0]$



Fe_ei200_align_sym3.sqw

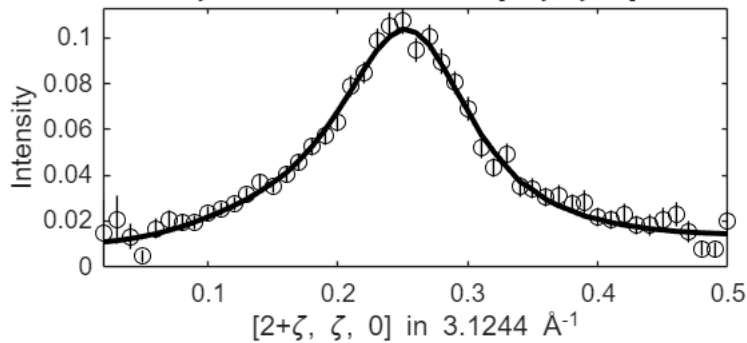
≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 130
 $\zeta = -0.005:0.01:0.505$ in $[2+\zeta, \zeta, 0]$



 **** En = 145±5 going to 155

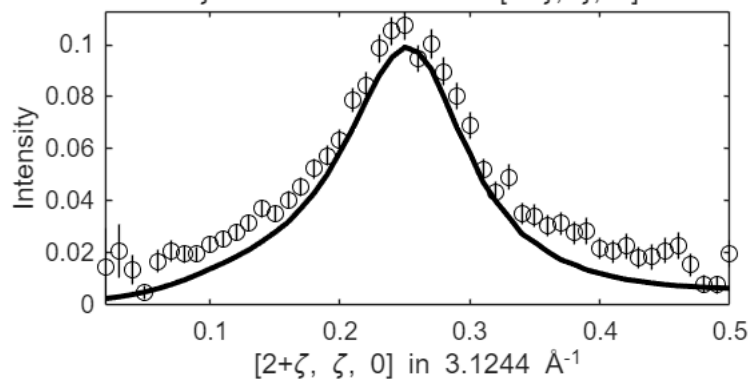
Fe_ei200_align_sym3.sqw

1.2 ± 0.066 ; $J_0 = 35 \pm 0.31$; $J_1 = 0 \pm 0$; $\gamma = 51.675 \pm 3.5$
 ≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 140
 $\zeta = -0.005:0.01:0.505$ in $[2+\zeta, \zeta, 0]$



Fe_ei200_align_sym3.sqw

≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 140 :
 $\zeta = -0.005:0.01:0.505$ in $[2+\zeta, \zeta, 0]$



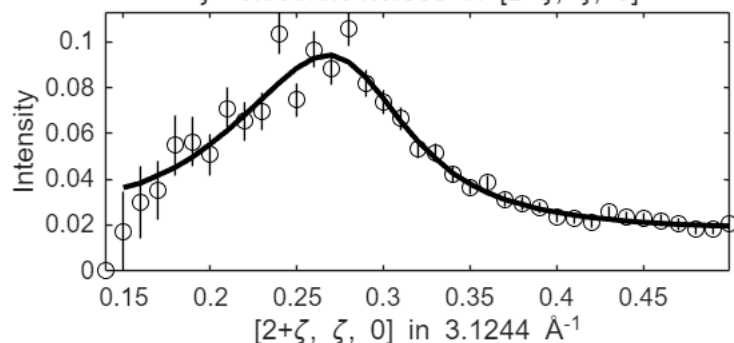
**** En = 155±5 going to 155

Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 1.9608

Fe_ei200_align_sym3.sqw

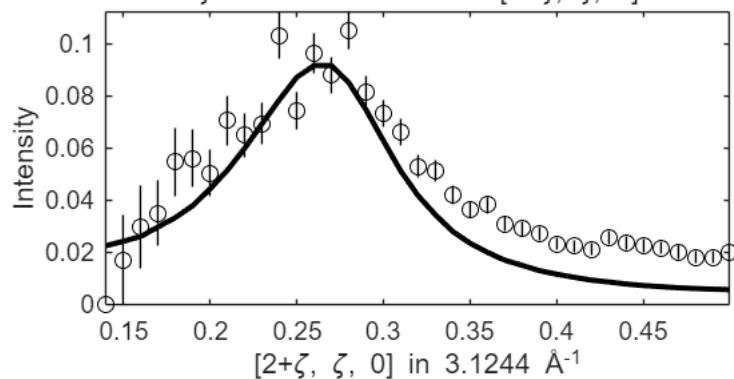
0.97 ± 0.05 ; $J_0 = 33 \pm 0.21$; $J_1 = 0 \pm 0$; $\gamma = 41.3538 \pm 2.1$

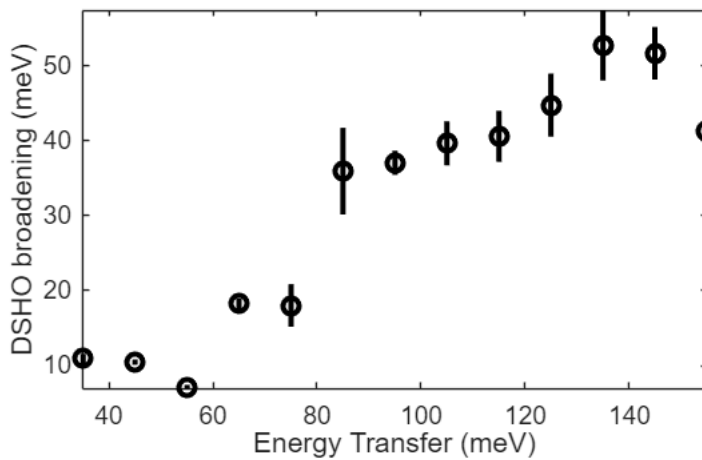
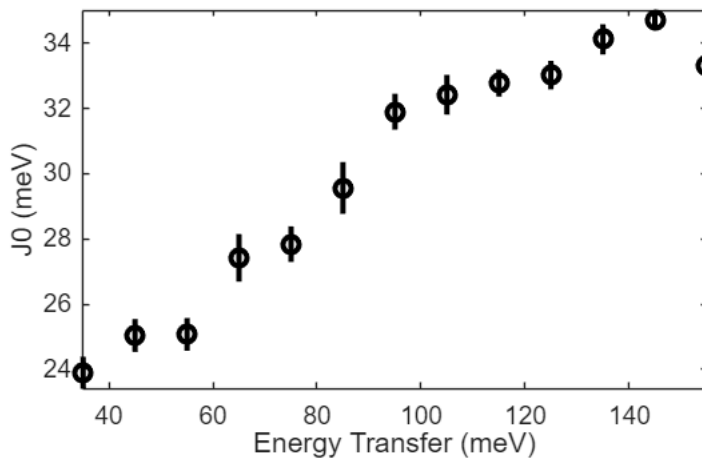
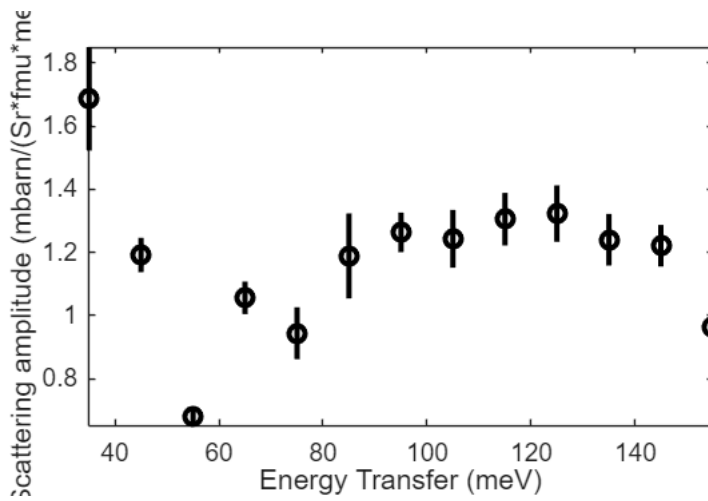
≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 150 :
 $\zeta = -0.005:0.01:0.505$ in $[2+\zeta, \zeta, 0]$



Fe_ei200_align_sym3.sqw

≤ 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 150 :
 $\zeta = -0.005:0.01:0.505$ in $[2+\zeta, \zeta, 0]$





```

cl = NG_FFCutsFits.(selectCuts);
NG_FFCutsFits.(selectCuts) = [cl{:}];

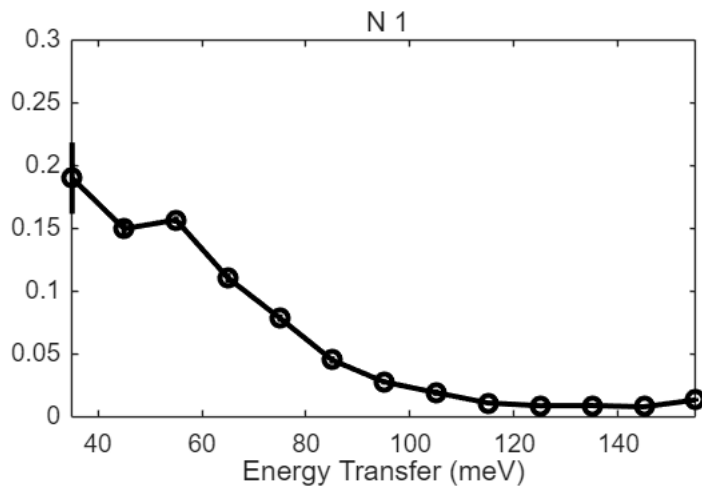
beep
beep
beep

all_fit_par = NG_FFCutsFits.([selectCuts, '_all_fit_par']);
bg_data = extract_bg_par(all_fit_par,[1,2]);

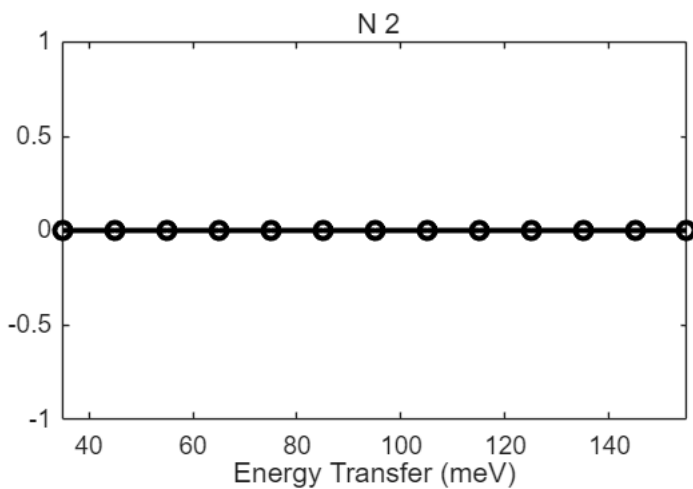
```

```
% original slope parameters
```

```
pd(bg_data(1));lx(all_fit_par(1).en,all_fit_par(end).en);ly 0 0.3; keep_figure;
```



```
pd(bg_data(2));lx(all_fit_par(1).en,all_fit_par(end).en);keep_figure
```



```
%pd(bg_data(3));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_figure;
%pd(bg_data(4));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_figure
```

```
doc = matlab.desktop.editor.getActive;
doc.save;
thisFile = 'e:\SHARE\Fe\Data\analysis\06_fit_with_AQ0\fit_cutFe_NG.mlx';
targ_file = fullfile(fileparts(thisFile),['fit_cutFe_NG_',selectCuts,'.pdf']);
export(thisFile,targ_file);
%close all;
%end
```

```
fit_src_struct = NG_FFCutsFits;
save(Cutfitfile,"fit_src_struct","-v7.3')
```